

Life history theory

Learning objectives

1. Understand how and why growth, survival, and fecundity trade off
2. Identify components of a life table
3. Calculate stage-specific mortality/survival

Review of last lecture: environmental variability and how to deal with that modeled population growth with seed bank

more into population structure - why might average something like seed mortality differ between species

Population structure

Terminology

- demography - structured populations
- vital rates - describe each stage *germination, survival to reproduction, fecundity*
- age and stage *related but useful in different circumstances*

stage examples: seeds, vegetative, reproductive

Why care about population structure?

- more realistic *that births and deaths are age/stage structured*

example: seedlings die in frost, reproductive adults don't

- different stages, different pressures *seeds-soil conditions, adults-competition*
- compare individuals within stage

Tradeoffs

limited amount of energy and resources - need to decide how to allocate to growth, survival, reproduction

most basic tradeoff is the r-K theory

r-K

- r species - fast growth r , many offspring, low investment
- K species - slow growth, near carrying capacity, few offspring, high investment
- **caveats** - binary instead of spectrum, generalization *example, trees like oaks that are long-lived but produce loads of acorns with low investment per acorn defy this theory*

Demographic triangle

- growth, stasis, fecundity

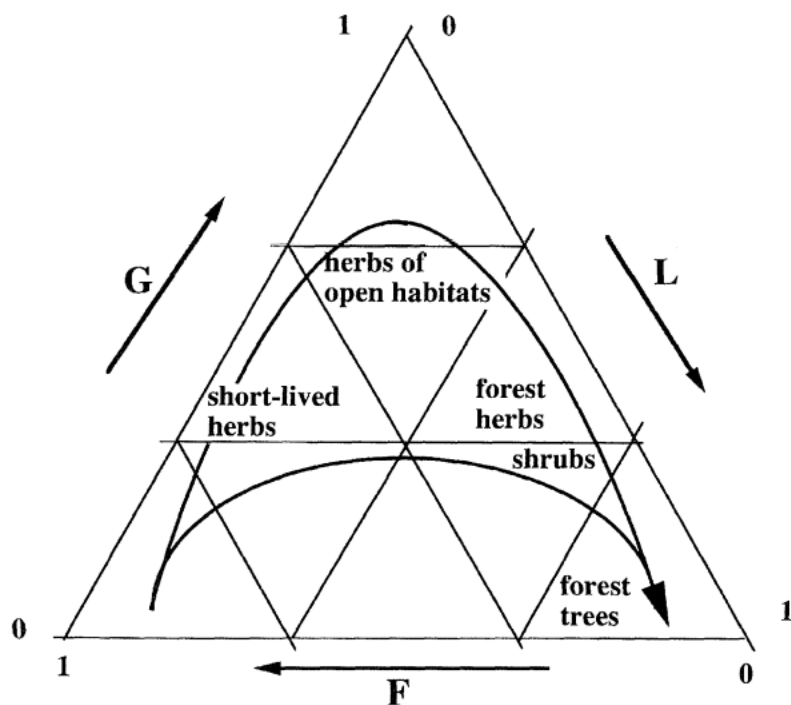


Figure 2. The successional trajectories suggested by the distribution of species in the demographic triangle in Fig. 1, after Silvertown & Franco (1993).

- can't optimize all three

Modeling life history stages

- life tables - rates at each stage
- abundance, survival, and fecundity

Stage	Description	Number Alive	Survival to Next Stage	Mortality	Fecundity
1	Seed	1000	0.20	0.80	0
2	Seedling	200	0.25	0.75	0
3	Adult	50	—	—	100

- convert to matrix *and perform analyses, which is what we'll do next week*